

DATA VISUALISATION PROJECT

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Who Pollutes, Who Pays?

A Multi-Dimensional Analysis of Global GHG Emissions,
Wealth, and Energy Access (2014–2023)

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1 Introduction and Research Question

Climate change is one of the most pressing challenges of our time, yet responsibility for greenhouse gas (GHG) emissions is unequal across nations. Public discourse often conflates low emissions with environmental virtue and high emissions with negligence, a framing that obscures the structural drivers of national carbon footprints, from wealth and industrialisation to energy access and historical development trajectories.

This paper presents *Who Pollutes, Who Pays?*, an interactive data visualisation system that deconstructs global GHG emissions across four analytical dimensions: geography, wealth, energy access, and measurement framing, drawing on OECD emissions data and three World Bank datasets covering economic development, energy systems, and population across 90 countries from 2014 to 2023.

The system is organised around four interlinked research questions:

1. **Where** are per-capita emissions highest, and how has the global distribution shifted from 2014 to 2023?
2. **Why** do some countries emit more than others, is wealth the primary driver, and which countries break the expected pattern?
3. **Are low-emission countries genuinely sustainable**, or does energy poverty explain their low footprint independently of deliberate decarbonisation choices?
4. **Does the choice of metric matter?** Can the same dataset produce opposite geopolitical narratives?

Together, these questions form a coherent analytical arc: from spatial description, through causal explanation, to critical reflection on the ethics of data visualisation.

2 Data

The system integrates four datasets spanning 90 countries over the period 2014–2023. The primary source is the **OECD GHG Emissions dataset**, which provides greenhouse gas emissions per capita in kg CO₂-equivalent per person, filtered to total emissions excluding Land Use, Land-Use Change and Forestry (LULUCF) and restricted to country-level observations by removing supranational aggregates such as the EU27 and OECD composites. Three **World Bank datasets** complement the OECD data: the Economy and Growth database supplies GDP per capita adjusted for purchasing power parity (PPP, current international \$), chosen over nominal GDP because PPP corrections make cross-country comparisons of living standards more meaningful; the Energy and Mining database provides renewable energy consumption as a share of total final energy consumption, access to electricity as a percentage of the population, and fossil fuel energy consumption share; and the Population dataset supplies total population counts used to derive absolute emission totals.

The three datasets from the World Bank were downloaded in their raw World Bank wide format (that is one row per country and indicator, years as columns) and transformed into a tidy structure of one row per country-year pair using a Python pipeline. The files were

unpivoted and filtered to the 2014–2023 window before being left-joined onto the OECD GHG dataset using the ISO-3 country code and year as composite keys. The OECD dataset served as the anchor for the join, meaning that only countries with at least one valid GHG observation are retained in the final panel. A World Bank seven-region classification was mapped onto each country manually, covering all 90 countries present in the GHG dataset. Absolute emission totals in megatonnes CO₂-equivalent were derived as the product of GHG per capita and total population, divided by 10⁹.

The integrated panel contains 789 country-year observations across 90 countries and 11 variables. Missing values are present but not imputed: GDP per capita is missing for 47 observations (6.0%), renewable energy share for 134 (17.0%), fossil fuel share for 75 (9.5%), and population for 9 (1.1%). Missingness is not random: it concentrates in small states and certain years for the energy variables, but given the analytical and visual nature of the system, listwise deletion per visualisation was preferred over imputation to avoid introducing artificial precision.

Several limitations bear acknowledgment. The OECD GHG dataset covers only 90 countries, predominantly high- and upper-middle-income economies, introducing a systematic selection bias: the countries with the most policy relevance for climate action are well-represented, but large parts of Sub-Saharan Africa and South Asia appear only partially or not at all. The renewable energy share variable includes traditional biomass, which can inflate the renewable share of low-income countries without reflecting a genuine clean-energy transition. Electricity access as a proxy for energy poverty is imperfect: a country may have near-universal grid access while still being economically poor. Finally, the analysis window ends at 2023 due to data availability constraints, and coverage thins noticeably after 2021 for several energy indicators as World Bank reporting lags increase.

3 Audience, Tasks and Design Rationale

The system targets an informed general audience: policy students, journalists, and engaged citizens with an interest in climate issues but no assumed statistical background. Design decisions consistently prioritised immediate readability over analytical complexity.

The system supports four analytical tasks in increasing depth: forming a spatial overview of global emissions; comparing countries and regions across income and energy dimensions; explaining why some countries emit more or less than expected; and critically reflecting on how metric choices shape the narratives that visualisations produce.

Each visual form was chosen to match its task directly. The Tableau dashboard provides the broad comparative layer: a choropleth map as spatial entry point and cross-filtering hub, an animated emissions race, a dynamic regional ranking, and a bubble scatter of renewable share against GDP per capita. The Python layer adds an animated scatter timeline tracking the wealth-emissions relationship over 2014–2023, and a white-hat/black-hat line chart pair demonstrating how axis and window choices produce opposite narratives from identical data. The JavaScript layer delivers a country profile card combining a gap timeline and a radar chart across four dimensions and a Transition Pressure Wheel encoding a composite climate-action pressure score as a radial stacked skyline across all countries.

4 System Description

The website presents the project as a guided public-facing dashboard rather than a simple collection of charts: users land on a main homepage, “Who Pollutes, Who Pays?”, which introduces the central research question, explains the data sources, and frames the analytical tasks before inviting exploration through linked visual modules. From this main dashboard, the narrative unfolds in a clear sequence: the Python page establishes methodological transparency through the white-hat/black-hat comparison, the Tableau dashboard supports broad cross-country filtering and comparison, the Country Profile page translates the data into a focused, easy-to-read national diagnosis, and the Pressure Wheel concludes with a more creative view that synthesizes climate pressure across multiple dimensions. In this way, the narrative is not separate from the interface: the links between pages are part of the storytelling itself, guiding the audience from context, to evidence, to comparison, and finally to interpretation.

5 Rejected Alternatives

During the development of the project, we considered several alternative directions before selecting greenhouse gas emissions as the final focus. One initial idea was to analyse the environmental cost of artificial intelligence by connecting model-level training footprints, such as GPU use, training time, energy demand and carbon emissions, with the broader electricity consumption of data centres. Although conceptually relevant, this option was rejected because the available data were too fragmented and often not directly verifiable: many AI companies and data centres do not disclose detailed energy, water or emissions information, while some model-level estimates rely on secondary reconstructions. A second option was to compare the efficiency of AI architectures, such as ResNet, VGG or MobileNet, in terms of energy use, CO₂ emissions and accuracy. However, this approach was narrower and less aligned with our aim of studying an environmental issue at a country or regional scale. For these reasons, greenhouse gas emissions were selected because they offered more transparent public data, broader geographical coverage, stronger links with socio-economic indicators, and a more reliable basis for both ethical and misleading visual interpretations.

6 Findings and Limitations

6.1 Key Findings

The analysis shows that climate responsibility cannot be reduced to a single metric, because countries occupy very different positions depending on whether we consider emissions per capita, total emissions, economic capacity, and energy context together. Across the project dataset, wealthier countries tend to emit more per person, yet this relationship is far from uniform: countries with similarly high GDP per capita display sharply different emissions profiles, suggesting that development alone does not determine climate pressure. The results also highlight the importance of scale, since some countries have very high per-capita emissions but relatively small total footprints, while others combine moderate per-capita values with

very large aggregate emissions due to population size. Regional comparison further changes interpretation, as a country may appear close to the global average while remaining highly atypical within its own region. Over time, many countries with comparable 2014 and 2023 observations show declining emissions intensity, although this trend is not universal and some cases still record increases. Finally, the findings suggest that low emissions should not automatically be read as evidence of successful transition, because in several lower-income contexts they coexist with lower electricity access and more limited economic capacity, while renewable-energy indicators must be interpreted cautiously due to weaker coverage in the most recent years.

6.2 White-Hat and Black-Hat Static Visualisation

To address the ethical dimension of the project, we developed a paired white-hat and black-hat static visualisation using the same data on average greenhouse gas emissions per capita in Europe and Central Asia. The white-hat version shows the full 2014–2021 trend with a zero-based y-axis, making the gradual long-term decline visible and preventing small year-to-year variations from being visually exaggerated. In this framing, the 2020 COVID-19 drop appears as part of a broader downward trajectory rather than as an isolated shock. By contrast, the black-hat version restricts the view to 2019–2021 and truncates the y-axis around the observed values. This magnifies the rebound from 2020 to 2021 and can misleadingly suggest that emissions are sharply increasing, even though the complete trend shows an overall decrease. The comparison illustrates that visualisation is not neutral: choices such as time window, axis scale, baseline and annotation can substantially shape the narrative extracted from the same data. For this reason, the project favours transparent framing, broader temporal context and explanatory annotations over visually dramatic but potentially misleading design choices.

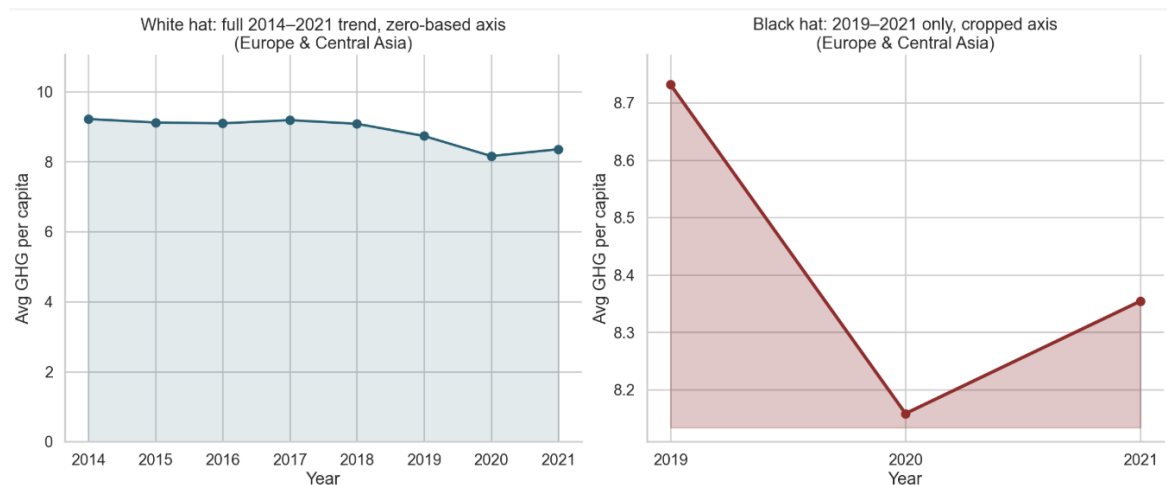


Figure 1: White-hat and black-hat representations of average GHG emissions per capita in Europe and Central Asia. The left panel shows the full 2014–2021 trend with a zero-based axis, while the right panel isolates 2019–2021 and crops the axis, visually exaggerating the post-COVID rebound.

6.3 Limitations and Future Work

Overall, the project should be interpreted as a descriptive and exploratory analysis rather than as a causal study. Its main limitation is the construction of the dataset, which is anchored

to the OECD per-capita GHG series and therefore does not provide a fully global sample. This creates a selection bias toward Europe and other data-rich economies, especially in the most recent years, where country coverage becomes thinner and geographically unbalanced. In addition, the integration of OECD and World Bank indicators introduces temporal inconsistencies, since not all variables are available for the same countries and years. Some indicators also require conceptual caution: the GHG measure excludes land-use and forestry emissions, while renewable energy consumption may include traditional biomass, potentially overstating the progress of some low-income countries toward modern clean energy systems. Finally, regional aggregation improves readability but can hide substantial differences between countries within the same region.

Future work could strengthen the project by expanding the emissions data beyond the OECD sample, especially for underrepresented regions such as Africa and South Asia. The analysis could also include both per-capita and absolute emissions, allowing users to compare fairness-oriented and total-impact perspectives. A more refined treatment of renewable energy, separating modern renewables from traditional biomass, would improve interpretability. Finally, adding further explanatory variables and a simple statistical layer, such as panel regression or country fixed effects, could help distinguish visual associations from more robust empirical relationships.

A AI Usage Appendix

A.1 Overview of AI Use

This project used Claude (Anthropic, claude.ai) as the primary AI assistant. AI was not used to generate findings, interpret results, or make analytical decisions.

Brainstorming: Claude was used as a thinking partner during the early design phase to stress-test visualisation choices against the data structure and the intended audience.

Data transformation: Claude assisted with the code for transforming raw World Bank CSV files into tidy country-year panels, in particular the unpivoting logic.

Debugging: Claude was used to locate errors in code blocks shared with the AI. All proposed fixes were reviewed and tested independently before being accepted.

Academic writing: Claude was used to refine phrasing and register. All content and arguments were drafted by the authors first; the AI was used only to improve clarity and remove redundancy.

A.2 JavaScript and Website Development

The JavaScript and website components required more use of AI assistance. The group had prior exposure to HTML, CSS, and JavaScript, though some patterns required revision

For the HTML and CSS of the website, Claude was consulted for specific layout patterns such as CSS grid, custom properties, and responsive breakpoints. AI suggestions for page structure and styling were reviewed, tested in the browser, and adjusted to match the design language of the project. The navigation logic, the hero layout, the data and findings sections, and the overall typographic system were all developed through an iterative dialogue between

the group's design decisions and AI-generated code proposals.

For the JavaScript layer, the scale of the experimentation is visible in the project repository, which contains ten prototype visualisations in an `old_vis` folder, developed and discarded before the two final views were selected. Claude was used to rapidly prototype D3.js patterns, evaluate whether a visual idea was technically feasible, and understand why a particular approach was not working. Many prototypes were initiated through a broad prompt describing the desired visual form and refined through many sequential interactions as the design evolved. The final Country Profile and Transition Pressure Wheel were both developed this way, with the group retaining full control over analytical decisions, interaction design, and the interpretation panels.